

Replicating and Optimizing the S&P 500 Buy-Write Strategy: A Market Microstructure Approach

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Introduction

A “covered call” or “buy-write” strategy involves holding a long position in an equity asset while simultaneously selling call options on that same asset. This generates premium income that buffers against market downturns, mitigating portfolio drawdowns during crises (Hibbert et al., 2020), although it caps the upside potential during bull markets. Covered-call ETFs are increasingly used by retail investors seeking income and downside protection in volatile markets (Morningstar, 2022). However, a misunderstanding of tracking error in these complex products may lead to severe mispricing of risk for these investors.

The Cboe S&P 500 BuyWrite Index (BXM) is a widely recognized benchmark for this strategy, and several Exchange Traded Funds (ETFs), such as the Invesco PBP ETF, are designed to track it. Unlike traditional equity index funds, derivative-based ETFs face significant challenges in accurately replicating their benchmark indices. The constant rolling of option contracts exposes the fund to market microstructure frictions, including bid-ask spreads, slippage, and execution timing mismatches (Pessina & Whaley, 2021). Furthermore, options pricing relies heavily on volatility estimates. Discrepancies between the market’s implied volatility (IV) and the realized actual volatility (RV) create parameter risks that can drastically affect the fund’s profitability.

While recent literature emphasizes the importance of downside protection strategies (Moenig, 2021), fewer studies quantify the role of execution microstructure in ETF-level replication error. This project contributes by isolating microstructure frictions as a measur-

able determinant of derivative-based tracking error. Specifically, this research aims to answer the following questions: How significantly do execution timing and microstructure rules affect the tracking error of a buy-write ETF? And, can dynamic option selection based on volatility spreads yield superior risk-adjusted returns compared to a static rollover strategy?

Hypothesis

This project tests two primary hypotheses: First, by strictly adhering to market microstructure rules (e.g., executing trades at the Volume-Weighted Average Price (VWAP) and using specific timestamped data for strike selection), the simulated model will significantly reduce tracking error against the actual Invesco PBP ETF compared to a simplified end-of-day execution model. Second, implementing a dynamic rollover filter based on the spread between Implied Volatility (IV) and Realized Volatility (RV) will yield a superior risk-adjusted return compared to the standard, static at-the-money (ATM) rollover strategy dictated by the BXM index.

Methods and Analysis

To ensure the project is realistic and achievable within the 120-hour UROP timeframe, a key control assumption will be applied: the underlying equity portfolio will be modeled to replicate the S&P 500 Total Return Index with zero tracking error. This allows the analytical focus to remain entirely on the options execution and optimization. To conduct this analysis, high-resolution historical Cboe S&P 500 options data (OPRA feed) and daily price data will be purchased directly from the Cboe Datashop. The acquisition cost for this data will be submitted as part of the UROP expense funding request. The research will be conducted in three progressive stages following an initial design phase:

Stage 1: Simple Thematic Replication. A baseline Python simulator will be constructed using a simplified approach. It will mechanically sell the most liquid, ATM call option on

the expiration Friday based on daily closing prices. The tracking error between this naive model and the actual PBP ETF will be calculated to establish a baseline.

Stage 2: Microstructure Fidelity. The engine will be upgraded to match the exact Cboe BXM methodology. Strike selection will be refined to target the closest strike at or above the 11:00 AM ET index value. To model realistic slippage, the sold option's price will be calculated using the 11:30 AM to 1:30 PM ET VWAP. Furthermore, the purchased OPRA feed data will be filtered to exclude delayed or canceled orders (codes A-H, f-t) to ensure data integrity.

Stage 3: Optimization via Volatility Spread. Expanding on the parameter risk models established by Henrard (2003) and Ahmad & Wilmott (2008), the engine will calculate the IV-RV spread. A dynamic switching algorithm will be tested: when IV is statistically overpriced relative to RV, the model will shift from the standard BXM strategy to alternative methodologies, such as selling 30-Delta (BXMD) options, to maximize expected profit.

Estimated Timeline: Weeks 1-2 will consist of an initial phase dedicated to an in-depth literature search, experiment design refinement, and the procurement of the Cboe Datashop data. Weeks 3-4 will focus on data cleaning and building the Stage 1 baseline simulator. Weeks 5-8 will be dedicated to the computationally intensive data engineering, filtering the OPRA feed, and implementing Stage 2 microstructure rules. Weeks 9-10 will focus on Stage 3 volatility optimization, data analysis, and finalizing the report.

Outcome

This project will produce a sophisticated backtesting algorithm capable of replicating complex derivative ETFs. Analytically, the research will quantify the hidden costs of market microstructure frictions in option roll strategies. The findings will provide valuable insights into quantitative portfolio management, helping retail investors better understand ETF risks, and will be disseminated through a comprehensive final report and a presentation at the University's Undergraduate Research Symposium.

References

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